Sales Performance Analysis

End-to-End Analysis Case Study: Python, Excel and Power BI

$118.7M	$16.9M	14.2%	700
Total sales	Total profit	Overall profit margin	Records, Sep 2013 to Dec 2014

Executive summary

This report analyses 700 sales records covering five countries, five customer segments and six products. The data was cleaned and explored in Python (Pandas), summarised in an Excel dashboard with pivot tables and slicers, and presented in a Power BI dashboard. The main findings are:

- **Government drives profit.** It earns $11.39M, about two thirds of all profit, at a 21.7% margin.
- **Enterprise loses money.** It sells $19.6M but makes a net loss of $0.61M (margin -3.1%). It is the only loss-making segment.
- **Discounts erode margin.** Profit margin falls from 21.9% on undiscounted sales to 9.1% in the High discount band.
- **Paseo is the top product** with $4.80M profit, about 28% of the total and 1.6 times the next product (VTT).
- **Sales peak in October and December.** October 2014 ($12.4M) and December 2014 ($12.0M) were the two best months.
- **The USA has the lowest margin** (12.0%) even though it has the highest sales ($25.0M).

Recommended actions (detail in Section 7): review Enterprise pricing and discount rules, cap High-band discounts, prioritise Paseo and the Government segment, investigate the USA margin gap, and plan stock and promotions ahead of the Q4 peak.

1. Objective and dataset

Objective. Understand which segments, products and countries drive sales and profit, find where the business loses margin, and give the stakeholders practical recommendations.

Dataset. 700 rows and 16 original columns: Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month Number, Month Name and Year. The data covers September 2013 to December 2014. One derived column, Margin % (Profit / Sales x 100), was added during cleaning, giving 17 columns in the cleaned file.

2. Method and tools

Stage	Tool	What was done
Clean and explore	Python (Pandas) in Jupyter	Structure checks, missing values, duplicates, consistency checks, derived columns
Analyse	Python (Pandas)	Group-by summaries by segment, country, product, year, month and discount band
Visualise (1)	Excel	Pivot tables, four pivot charts and four slicers on a Dashboard sheet
Visualise (2)	Power BI Desktop	Four visuals and four slicers (Segment, Product, Country, Year) on one page

3. Data cleaning

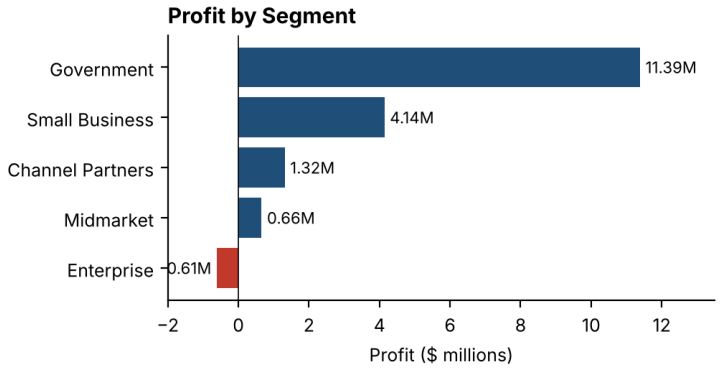
The cleaning steps below were run in the Python notebook, and the cleaned data was exported as clean_sales.csv.

Check	Result	Action
Shape	700 rows x 16 columns	None needed
Column names	Stray spaces possible	Whitespace stripped from all names
Missing values	53 blanks, all in Discount Band	These rows all have zero discount, so blanks were filled with "None"
Duplicate rows	0	None needed
Gross Sales - Discounts = Sales	0 mismatches	Data is internally consistent
Sales - COGS = Profit	0 mismatches	Data is internally consistent
Derived column	Margin % added	Profit / Sales x 100, rounded to 2 decimals

Discount Band after cleaning: High 245 rows, Medium 242, Low 160, None 53.

4. Findings

4.1 Profit by segment

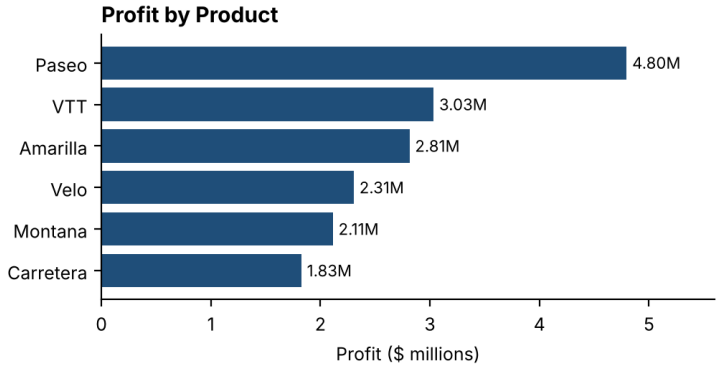


Government earns **$11.39M** profit on $52.5M sales (21.7% margin). Small Business is second with $4.14M but a thin 9.8% margin. Channel Partners and Midmarket are small but efficient, at 73.1% and 27.7% margins.

Enterprise makes a loss of $0.61M on $19.6M of sales. It has 58 loss-making rows, and every one of them is an Enterprise sale.

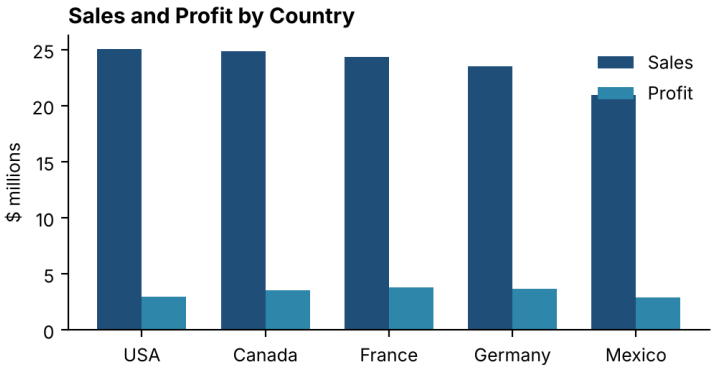
Segment	Sales ($M)	Profit ($M)	Margin
Government	52.50	11.39	21.7%
Small Business	42.43	4.14	9.8%
Channel Partners	1.80	1.32	73.1%
Midmarket	2.38	0.66	27.7%
Enterprise	**19.61**	**-0.61**	**-3.1%**

4.2 Profit by product



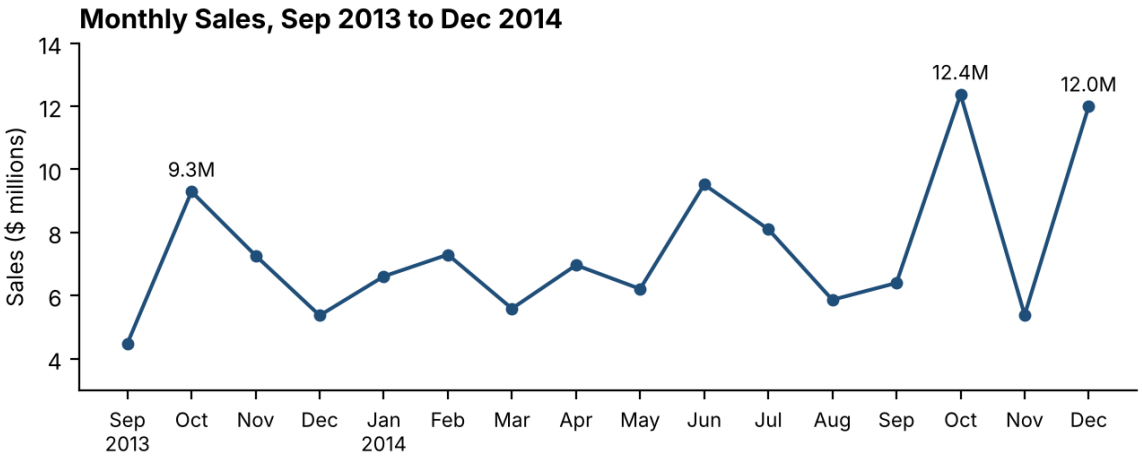
Paseo is the clear leader with $4.80M profit (about 28% of the total) on 338,240 units, about double the units of any other product. VTT ($3.03M) and Amarilla ($2.81M) follow. Carretera ($1.83M) and Montana ($2.11M) are the weakest. Margins are similar across products (12.6% to 15.9%), so the gap comes from volume, not price.

4.3 Sales and profit by country



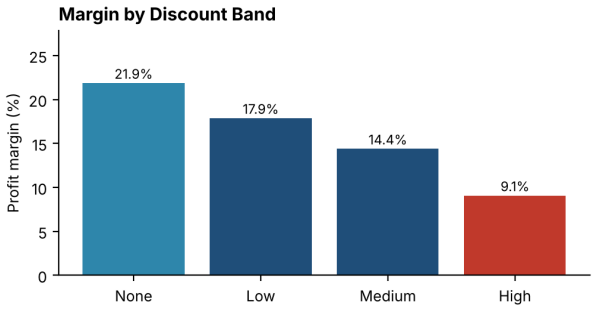
Sales are spread evenly across the five countries ($20.9M to $25.0M). Profit is not: **France ($3.78M) and Germany ($3.68M)** lead, with margins of 15.5% and 15.7%. The **USA has the highest sales but the lowest margin (12.0%)**, and Mexico has the lowest profit ($2.91M).

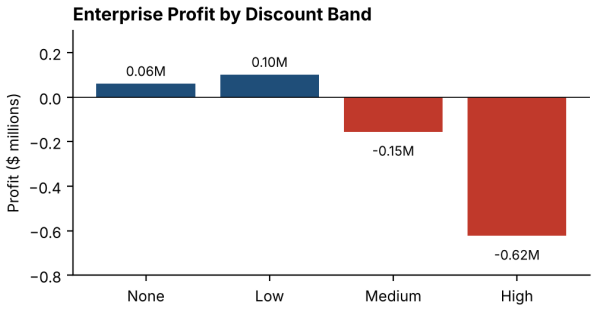
4.4 Monthly sales trend



Sales are seasonal. The two highest months are **October 2014 ($12.4M)** and **December 2014 ($12.0M)**. September and November 2014 are low in between ($6.4M and $5.4M). The same Q4 pattern appears in 2013, with October 2013 at $9.3M. Sales for 2014 total $92.3M against $26.4M for 2013, but 2013 contains only four months (Sep to Dec), so the two years should not be compared directly.

4.5 Discounts and the Enterprise loss





Margin falls steadily as discounts grow: 21.9% with no discount, 17.9% (Low), 14.4% (Medium) and 9.1% (High). In the Enterprise segment, High-band discounts alone lose $0.62M and Medium-band discounts lose $0.15M. Only Low-band and undiscounted Enterprise sales are profitable.

4.6 Why Enterprise loses money

The Enterprise price is only about 4% above its unit cost, and Enterprise receives the same average discount of about 7% as every other segment. A discount of that size wipes out the thin margin.

Segment	Price / unit	Cost / unit	Avg discount	Net price	Profit / unit
Channel Partners	12.00	3.00	6.95%	11.17	+8.17
Enterprise	**125.00**	**120.00**	**6.92%**	**116.35**	**-3.65**
Government	119.83	87.36	6.91%	111.55	+24.19
Midmarket	15.00	10.00	7.77%	13.83	+3.83
Small Business	300.00	250.00	7.65%	277.05	+27.05

Net price = price x (1 - average discount). Enterprise nets about $116.35 per unit against a cost of $120, a loss of about $3.65 per unit. Every other segment stays positive.

5. Dashboards

Excel dashboard. Four pivot charts (Profit by Product, Sales and Profit by Country, Monthly Sales Trend, Profit by Segment) and four slicers (Segment, Country, Product, Year), all connected to every pivot table.

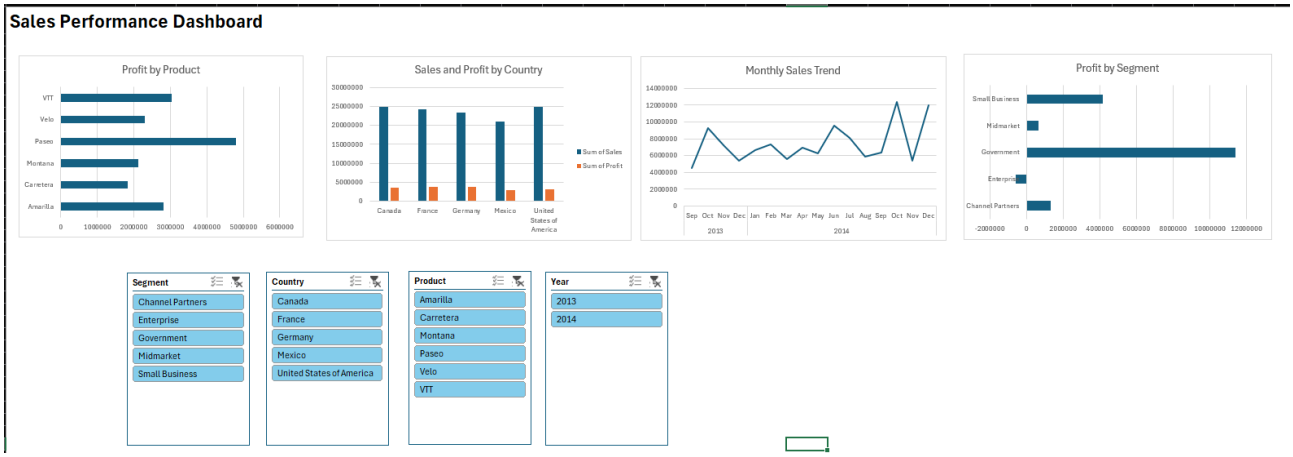


Figure: Excel Dashboard sheet with pivot charts and slicers.

Power BI dashboard. The same four charts and four slicers built in Power BI Desktop from clean_sales.csv. Slicers filter every visual on the page, for example selecting Enterprise in the Segment slicer shows the negative profit across the page.

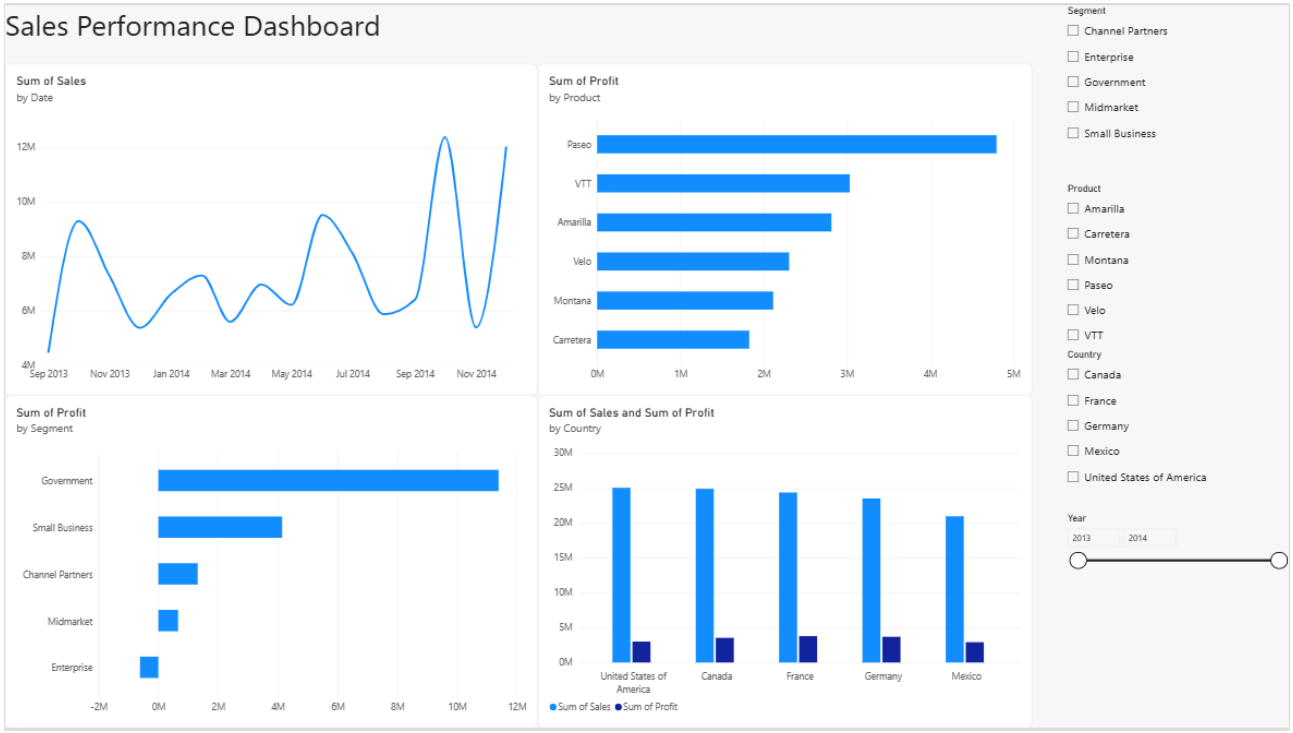


Figure: Power BI dashboard (Sales_Dashboard / sales.pbix).

6. How to use the dashboards

- Click a value in any slicer (Segment, Country, Product or Year) to filter all charts at once. Click it again to clear the filter.
- Select **Enterprise** in the Segment slicer to see the loss. Combine it with a Product or Country to find where it is worst.
- Select a single **Year** to compare 2013 and 2014, remembering that 2013 holds only four months.

7. Recommendations

- **Fix Enterprise pricing and discounts.** Raise the Enterprise price or lower the discount for this segment. At the current cost of $120 per unit, the segment needs a net price above that to break even. Cap High-band discounts first, because they cause $0.62M of the loss.
- **Set discount rules by segment.** A flat discount of about 7% is applied to every segment. Enterprise has no room for it, and Small Business (9.8% margin) has little.
- **Invest in Paseo and the Government segment.** Together they are the largest and most reliable sources of profit. Protect their stock and customer relationships.
- **Grow Channel Partners and Midmarket.** Both earn high margins (73.1% and 27.7%) from a small sales base, so they are the best candidates for growth.
- **Investigate the USA margin gap.** The USA has the most sales but the lowest margin (12.0%) against 15.5% to 15.7% in France and Germany. Check segment and product mix, and discounts, in that market.
- **Plan for the Q4 peak.** Build stock and schedule promotions before October and December, and avoid heavy discounting in the months that already sell well.

8. Limitations

- The data covers only September to December 2013 but all of 2014, so year-over-year growth cannot be measured reliably.
- The dataset has no customer-level or cost-breakdown detail, so the cause of the USA margin gap needs further data.
- Averages by discount band mix segments. For example, Channel Partners (high margin) and Enterprise (low margin) have different shares of each band.

9. Conclusion

The business is profitable overall (14.2% margin), but profit is concentrated in the Government segment and the Paseo product, and discounting is cutting margin everywhere. Enterprise is the one segment where the current price and discount combination loses money. Fixing Enterprise pricing and tightening discount rules gives the largest and fastest improvement.